A Feature Prefusion and Mask-Guided Network for Camera Decoration Defect Detection

Hui Wang, Yuqian Zhao, Fan Zhang, Gui Gui, *Member, IEEE*, Qiwu Luo, *Senior Member, IEEE*, Chunhua Yang, *Fellow, IEEE*, and Weihua Gui, *Member, IEEE*

***Abstract*— Camera decoration is an important part of smartphone. To achieve fully automated production, a dependable, efficient, and automatic method is required for camera decoration surface defect detection. This article presents a detection scheme based on computer vision to improve the efficiency of screening defective products. Since there is no available dataset for method designing in camera decoration field, we establish a camera decoration defect dataset CD3 including 9417 samples with four types of defects. To increase sample size and alleviate category imbalance of CD3, we provide a dataset enhancing framework including a defect copy method and a background reuse method to generate an enhanced dataset CD3_En containing 39 649 samples. Besides, a feature fusion and mask-guided network (FMN) including a feature prefusion (FPF) module and a multistage fusion (MSF) module is proposed to screen the defective products. The FPF is constructed by receptive field blocks (RFBs) and information diffusions (IDs), and it can achieve data volume reduction and context enhancement after being embedded between the BoneNet and Neck. The MSF is used as the Neck to realize a two-step feature fusion for predicting the bounding boxes of defects and their masks. The experimental results on the CD3_En dataset demonstrate the superiority of the proposed method compared with other 11 classic object detection methods.**

***Index Terms*— Camera decoration, context enhancement, dataset enhancing, defect detection, mask guidance.**

I. Introduction

CAMERA decoration as a key component of smartphone is produced in billions annually. During the camera decoration production process, insufficient technology may induce a variety of surface defects, such as grain, scratch, crush, and radium carving injury (RC), as shown in Fig. 1. To ensure product quality, camera decorations need to undergo strict inspection to screen out defective samples, which is mainly implemented manually so far. Manual quality inspection has the deficiencies of strong subjectivity, high uncertainty, and low efficiency, which cannot meet the requirements of modern industry for high-speed and high-precision inspection. A fast and accurate method to accomplish quality inspection is needed to increase the efficiency of production and decrease the reliance on manual labor. However, to our knowledge, there is still no relevant research so far in the field of defect detection for camera decorations.

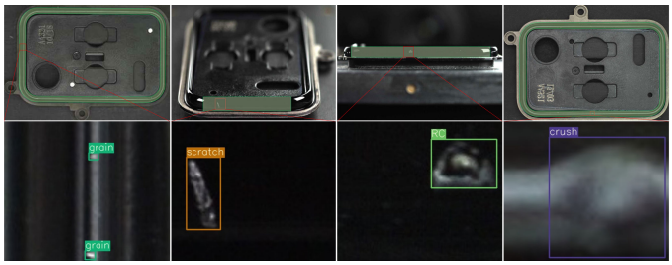


Fig. 1. Examples of camera decoration surface defects. The first row is camera decoration defect images photographed by industrial camera, and the second row is the amplified images of the defective regions marked by red boxes in the first row. The regions covered by the green masks are the inspection areas.

In many other industry fields, researchers have proposed many quality inspection methods based on computer vision to compensate for the shortcomings of manual defect detection. Some traditional methods, such as grayscale covariance matrix, mixed Gaussian Markov random field (GMRF), uniform local binary pattern operators, and adaptive morphological reconstruction, have been studied and applied in some industrial fields and overcome the deficiencies of manual quality inspection to some extent. However, these methods mainly rely on manual prior knowledge and are therefore difficult to obtain accurate and universal defect patterns, leading to poor detection performance and limited applications.

Recently, deep-learning-based detection methods without introducing manual prior have made impressive advancements by exploiting powerful feature learning ability of neural networks. Compared with traditional methods, neural networks use gradient descent algorithms [1] to automatically find the most suitable parameters for generating defect feature maps. Manual prior knowledge is no longer indispensable in setting the parameters of algorithms. Therefore, defect detection methods based on neural networks have better universality as long as their sample data are provided. The deep-learning-based methods have been applied in many fields, such as rail

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Hui Wang and Fan Zhang are with the School of Automation, Central South University, Changsha 410083, China (e-mail: 224601027@csu.edu.cn; zhangfan219@csu.edu.cn).

Yuqian Zhao, Gui Gui, Qiwu Luo, Chunhua Yang, and Weihua Gui are with the School of Automation and the Key Laboratory of Industrial Intelligence and Systems, Ministry of Education, Central South University, Changsha 410083, China (e-mail: zyq@csu.edu.cn; guigui@csu.edu.cn; luoqiwu@csu.edu.cn; ychh@csu.edu.cn; gwh@csu.edu.cn).

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wheel tread defects [2], strip steel surface defects [3], printed circuit board defects [4], fabric defects [5], and tire defects [6], and can be divided into supervised and unsupervised methods depended on whether manual labels are used. Generally, the supervised methods have better performance in defect detection and are widely used in various industry fields' quality inspection [7], [8], [9], [10].

For training neural networks, a large emphasis needs to be placed on building the defect detection dataset. A terrible sampled dataset with low-quality images and labels may return poor detection performance even if the state-of-the-art model is used [11]. High-quality dataset for object detection in real-life scene can be obtained easily, while high-quality one for defect detection in industry scene is difficult to obtain for it requires more time and labor costs to collect and label sample data manually. As we know, only several defect detection datasets have been built in industry, such as NEU-DET [12] for hot strip steel, RSDDs [13] for rail surface, KolektorSDD [14] for electronic commutators, and Oliena [15] for glass bottles. Unfortunately, the above datasets are often small in size and may have class imbalance issues, resulting in difficultly generating a well-performing model, and to our knowledge, there is no available dataset so far for defect detection of camera decorations.

To realize surface defect detection for camera decorations, we establish a camera decoration defect dataset (named as CD3) with 7475 images in the training set and 1942 ones in the test set as there is no such dataset before. To improve the quality of the dataset, a dataset enhancing framework including a defect object copy method and background reuse method is proposed to expand the dataset and address category imbalance issue, and a new enhanced dataset CD3_En is generated containing 37 707 images in the training set and 1942 ones in the test set. To solve the problem of camera decoration quality inspection, we treat defect detection as an object detection problem and propose a feature prefusion (FPF) and mask-guided network (FMN) to detect surface defect. The FMN mainly includes an FPF module and a multistage fusion (MSF) module. The former is inserted between the BoneNet and the Neck to reduce the feature map channel and diffuse the multidimensional information between feature maps, and the latter is the Neck that contains two feature fusion structures to realize mask prediction and object detection. We test the proposed FMN on our established dataset and select some challenging scenes to compare the detection results of different methods.

Our main contributions can be summarized as follows.

1) A dataset is established with 9417 well-labeled samples for camera decoration defect detection, and a dataset enhancing framework including the defect copy and background reuse is proposed to address the class imbalance problem and increase the enrichment of dataset.
2) An effective defect detection framework FMN is proposed to realize the camera decoration defect detection, and two novel designed modules, FPF and MSF, are integrated into FMN to achieve context enhancement and multiscale feature fusion guided with masks, respectively.
3) Extensive experiments demonstrate the superiority of the FMN on camera decoration defect detection compared with other 11 classic defect detection methods.

II. Related Work

A. Dataset Expansion Method

Due to the high cost of collecting and labeling data in the industry field, it is difficult to establish a large dataset which is necessary for training neural networks. Moreover, different defect categories with big number gaps bring about the class imbalance problem. To address these issues, some datasets enhancing framework are proposed to generate high-quality dataset.

A common way to enhance dataset is copying defect targets and pasting them into the background. By controlling the copy and paste times of different types of defects, more labeled targets are generated and the number of labeled objects for different categories is balanced. Kisantal et al. [16] proposed a dataset enhancing framework based on target replication within an image. For an image with defects, subgraphs extracted with defects are inserted into the image at a random position. However, the complexity of the dataset does not improve significantly due to the limited defect replication which is operated within a single image. Dvornik et al. [17] proposed a dataset enhancing framework by synthesizing object instances and backgrounds crossing images. To guarantee the synthesizing quality, a new neural network is designed to generate the context which is used to filter the instance to decrease the semantic conflict between the instance and background. It is worth noting that the algorithm has a high time complexity due to the neural networks used for context generation.

Different from the copy-based methods, generation-based methods form new samples by imitating the information structure of existing samples in the dataset. Hinton et al. [18] used restricted Boltzmann machines (RBMs) based on the energy probability distribution to generate images with/without labels. Shin et al. [19] adopted the image-to-image translation conditional GAN to expand the medical image dataset while protecting the privacy of patients. Niu et al. [20] proposed a deep-learning-based sample generation method called the surface defect generation adversarial network (SDGAN) which has two generators. One generator is used to generate defect samples from normal samples, and the other one is used to restore defect samples to normal samples. With abundant data and careful tuning, high-quality images can be generated by the above GAN-type methods. However, insufficient defect datasets, poor diversity, and limited time make it unsuitable for generating industrial defect images.

B. Defect Detection Method

Many methods have been explored on defect detection in the industrial field. For traditional methods, defect detection can generally be seen as an analysis on color, contour, and texture. Researchers classified the techniques into four categories for extracting color, contour, and texture features: statistical, structural, filter-based, and model-based methods. Capizzi et al. [21] proposed a gray level concurrence matrix

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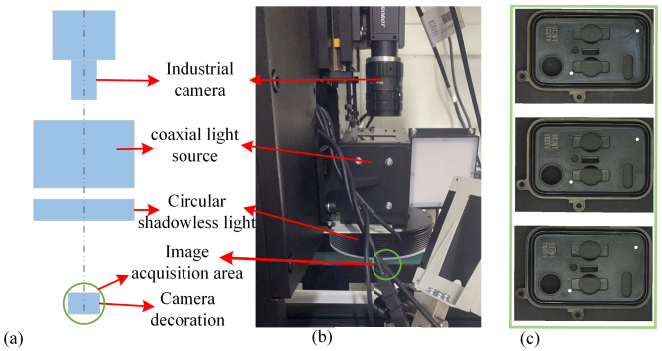


Fig. 2. Equipment for camera decoration image collection (partial). (a) Equipment diagram. (b) Equipment. (c) Image of front.

(GLCM) with a radial basis probability neural network for detecting defects on fruits. Cohen et al. [22] used the GMRF to model defect-free textures on fabric images for defect detection. Due to poor universality and high usage complexity, traditional methods are difficult to use widely.

Different from the traditional methods, neural networks that use data instead of manual prior to optimize the parameters perform better in defect detection problems. The defect detection technology based on deep learning can be divided into two categories: network based on GAN and network based on object detection framework. The basic concept of the GAN-based network is to establish a reconstruction network to fix the defect area. Donahue et al. [23] first applied AnoGAN to defect detection in electronic components in 2019. Zhao et al. [24] proposed the combination of GAN and an automatic encoder for industrial product defect detection.

The defect detection methods based on object detection framework regard defects as special targets to be inspected. Object detection methods based on neural networks can be divided into a single-stage method based on regression and a two-stage method based on candidate box generation and classification. Redmon et al. [25] proposed a representative single-stage object detection algorithm, You Only Look Once (YOLO), which inherits from OverFeature. The speed advantage makes it a leader in end-to-end target detection. Liu et al. [26] proposed the single-shot detection (SSD) method, which is widely used in industrial fields [27]. The two-stage target detection algorithm originated from the rich feature convolutional neural network (R-CNN). Ren et al. [28] proposed the Faster R-CNN, which adds the RPN to the backbone network structure and sets multiscale anchors according to the established rules. Chen et al. [29] proposed an improved Faster R-CNN for fabric defect detection based on a Gabor filter with genetic algorithm optimization.

III. Methodology

A. Overview

Surface defects on camera decorations main include four types: grain, scratch, crush, and RC. Since there is no dataset in the field of camera decoration defect detection, we first collect 1320 camera decoration defect images and establish a defect dataset named CD3 in this article. Similar to other industrial defect datasets, CD3 also faces issues such as insufficient data volume and imbalanced categories. To address these issues, we devise a dataset enhancing framework including a defect object copy method and a background reuse method to construct an enhanced dataset CD3_En. Besides, to solve the problem of relying on manual labor for camera decoration quality inspection, we propose an FPF and mask-guided network based on enhanced CD3 to screen out defective products.

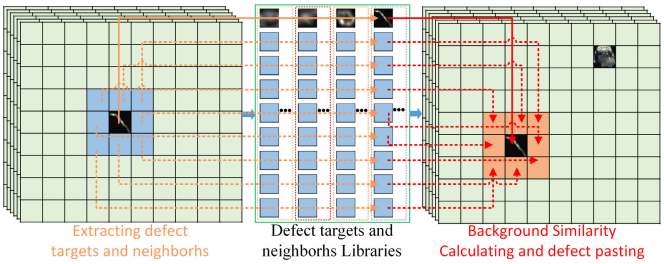


Fig. 3. Principle of the defect copy method (four defect libraries).

B. Dataset and Expansion

1) Construction of CD3: Camera decoration image collection equipment is shown in Fig. 2. We use it to collect images from 170 camera decorations and label samples with the software named *labelme*. For each camera decoration, 17 images are token by the industrial cameras to cover all the inspection areas. We collect a total of 1320 defect images with defects labeled manually. For an image captured by the industrial camera, the inspection areas only accounts for a small portion. To accelerate training and inference, the inspection area is cut out and participates in detecting. It is worth noting that some defect images contain multiple inspection areas and some inspection areas are defect-free. After the inspection areas are cut out, we get 1514 inspection area images which are divided into training and test sets based on the ratio of 8:2.

The calculating burden is greatly decreased by cutting the inspection areas from defect images. However, the aspect ratios of the inspection areas images are extreme and are therefore not friendly to neural networks. Hence, we process each inspection area image with a sliding window to generate subimages with an aspect ratio close to 1, and the stride step is set to half of the window's size. When processing the inspection area images in training dataset, all the defect-free subimages and original inspection images are discarded. While in test dataset, all the subimages are kept except for original inspection area images. By this way, we construct a new dataset, CD3, with 7475 images in the training set and 1942 ones in the test set. However, there are two obvious shortcomings in this dataset. First, serious category imbalance due to different probabilities of four different types of defects occurring. Second, low complexity since most subimages obtained from the inspection areas have been discarded for their lack of defects.

2) Defect Copy Method: To balance four different defect categories, a defect copy method crossing images is proposed with an environment similarity measuring function to achieve high-quality dataset expansion and erase the semantic conflict, as shown in Fig. 3. Compared with the neural network which is used to generate a context for filtering defect targets, the environmental similarity measuring function has

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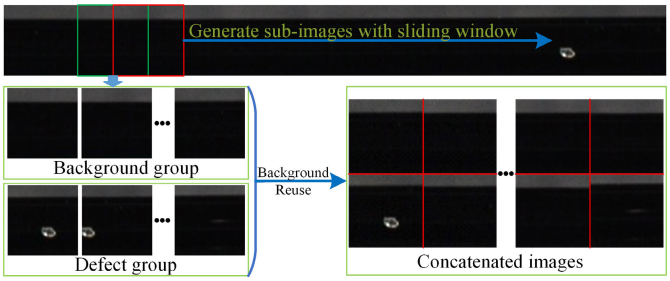


Fig. 4. Process of background reuse method.

a lower computational complexity and is easy to achieve. In our defect copy method, we construct four different defect libraries by category according to the defect targets and their eight neighborhoods extracted on the manual labels. For each inspection area image, some defect targets will be pasted into their corresponding standby positions to alleviate the category imbalance issues. These defect targets are randomly chosen from the defect libraries, and the standby position for each defect target is randomly generated within the inspection area. To guarantee contextual semantic consistency after pasting the defects into their standby positions, we need to calculate environment similarity ES between the source environment sE and its corresponding target environment tE, and ES is defined as

$$\mathrm{ES} = \frac{1}{8}\sum_{i=1}^{8} \mathrm{SSIM}(sE_i, tE_i) \tag{1}$$

where sE_i is the ith neighborhood around the source position where the defect target is extracted from, tE_i is the ith neighborhood around the standby position, and SSIM is the structural similarity formula which describes the image similarity from luminance similarity, contrast similarity, and structure similarity [30]. The SSIM is defined as

$$\mathrm{SSIM}(sE, tE) = \frac{(2\mu_{sE}\mu_{tE} + c_1^2)(2\sigma_{sEtE} + c_2^2)}{(\mu_{sE}^2 + \mu_{tE}^2 + c_1^2)(\sigma_{sE}^2 + \sigma_{tE}^2 + c_2^2)} \tag{2}$$

where μ_{sE} and μ_{tE} are the averages of sE and tE, respectively; σ_{sE}^2 and σ_{tE}^2 are the variances of sE and tE, respectively; σ_{sEtE} is the covariance between sE and tE; and c_1 and c_2 are generally set to 0.01 and 0.03, respectively. When ES ranging from 0 to 1 is smaller than 0.8, the standby position for pasting is regarded as unsuitable for defect target, which otherwise will lead to semantic conflict and decrease the quality of the dataset. Only when ES is above 0.8, the defect target will be pasted into the standby position. The category imbalance issue in CD3 will be addressed by controlling the number of defect targets chosen from each library.

3) Background Reuse Method: During the subimage generation process by the sliding windows, all the subimages which do not contain defects are abandoned, which decreases the enrichment of the dataset CD3. Therefore, we proposed a background reuse method to add subimages only containing background to the dataset, as shown in Fig. 4. A sliding window with a certain size is used to crop the inspection area to generate subimages. All the subimages are divided into two groups according to containing defects or not. We select three subimages from the background group and one subimage from the defect group to concatenate for generating a new image, and these subimages' positions in the new image are generated randomly. It is worth noting that both the generated new image and the subimage from the defect group must be included in the dataset for training.

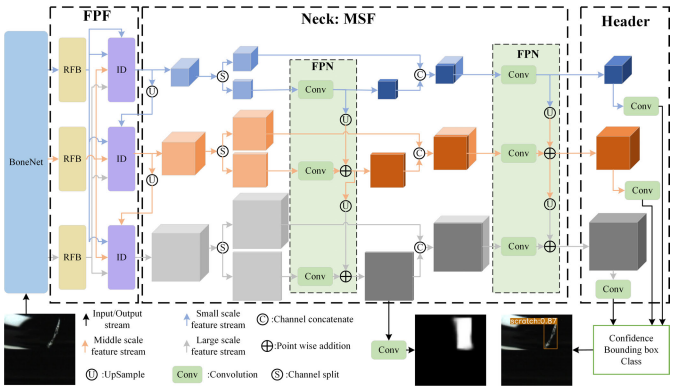


Fig. 5. Overview of the proposed FMN framework. The RFB, ID, and FPN are the receptive field block, the information diffusion module, and the feature pyramid network, respectively.

4) Construction of CD3_En: To address the issues of insufficient data and imbalanced categories existing in CD3, the proposed dataset enhancing framework including a defect copy method and a background reuse method is applied on the collected camera decoration defect images to present an enhanced dataset CD3_En. During the process of constructing CD3_En, the defect copy method is first used after the inspection area images being cut out from the defect images. By controlling the copy and paste times of different types of defects, more labeled defects are generated and the number of labeled defects for different categories is balanced. Then, the background reuse method is adopted during subimage generation process to concatenate the defect-free subimages with defective subimages. Therefore, a large amount of background is preserved which enriches the diversity of dataset. And to ensure the effectiveness of verification, the dataset enhancing framework is only applied on the training set.

C. Feature Prefusion and Mask-Guided Network (FMN)

We adopt the framework based on object detection instead of that based on generative adversarial to construct an FMN for defect detection. The proposed FMN framework includes four components: BoneNet for feature extraction, the proposed FPF for information permeation, the proposed MSF for feature fusion with mask guidance, and Header for defect detection, as shown in Fig. 5.

To ensure the accuracy of defect detection, we adopt DarkNet53-CSP [31] as the BoneNet to extract the multiscale feature from input images. The DarkNet53-CSP is a classic feature extraction network and has been widely used in object detection algorithms including YOLOx [32] and Scaled-YOLOv4(SYOLOv4) [31]. Besides, the Header adopt in FMN has the same structure as Header of SYOLOv4 composed of three convolution layers.

When an image is sent to the BoneNet, three feature maps of different scales containing different information are

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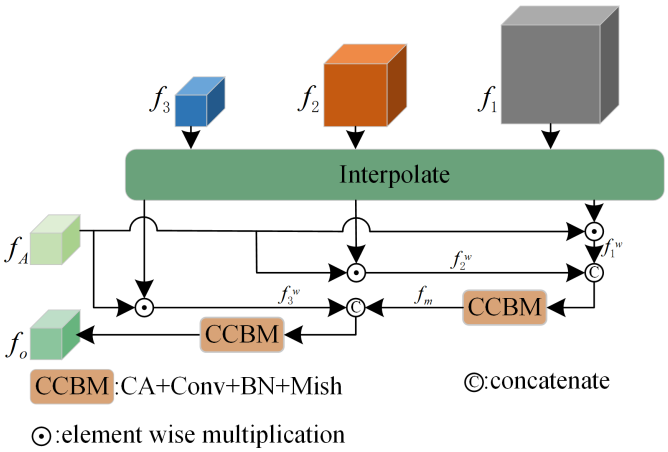


Fig. 6. Structure of ID module. CA, Conv, BN, and Mish represent the channel attention module, the convolution layer, the batch normalization layer, and the Mish activation function, respectively.

generated and sent to the FPF. The FPF reduces the data volume and enhances the representation ability of multiscale feature maps. The MSF adopts a two-stage feature fusion to process the multiscale feature maps coming from the FPF for defect mask prediction and feature map fusion. Finally, the Header which adopts the fused feature maps as input yields the predicting results including confidence, category, and bounding box.

1) FPF Module: To improve the inference speed of the subsequence network and the accuracy of the prediction results, we propose the FPF module to realize the data volume reduction and information permeation for context enhancement among multiscale feature maps. The FPF includes two parts: the RFB [33] and the ID module. The RFB is used to expand the receptive field of the feature maps coming from the BoneNet, which is realized by the dilated convolution with multiscale dilation and multiscale convolution kernels. The feature maps processed by RFB are separately convoluted by four branches. Each branch adopts the dilation and convolution kernel with certain scale to provide a receptive field with certain scale. The feature maps coming from the four branches are concatenated in the channel dimension and are then processed with a convolution layer to generate results with certain channel number. The expansion of the receptive field makes feature maps contain more context and increase the information density within the feature maps. By controlling the hyperparameter of RFB, the channel of feature maps generated by RFB is reduced to 128 which therefore decreases the data volume of feature maps. Meanwhile, the information contained in feature maps is also guaranteed as information density has been improved by the dilated convolution applied for receptive field expansion.

In the FPF module, each ID module adopts attention feature maps f_A and three different-scale feature maps f_1, f_2, and f_3 are generated by three RFBs to realize the information permeation between multiscale feature maps. Three different-scale feature maps are processed with an interpolate function to realize shape normalization by referencing the shape of attention feature maps, as shown in Fig. 6. Three interpolated feature maps are weighted by the attention map to generate the prior guided feature maps f_1^w, f_2^w, and f_3^w. To realize the context enhancement of a certain scale, the feature maps f_1^w and f_2^w are concatenated in the channel dimension, followed by the CCBM to generate feature maps f_m realizing ID and context enhancement for the first time. The CCBM includes four components: the channel attention module [34], convolution layer, batch normalization layer, and mish activation function. The feature maps f_3^w and f_m are also concatenated in the channel dimension, and the achieved result is then handled by the CCBM to output the final feature maps f_o. The certain scale context contained in the output feature maps has been enhanced by the information diffused from multiscale context. Significantly, the attention maps used as the prior guidance come from the RFB or another ID, as shown in FPF of Fig. 5. Each ID is responsible for the context enhancement of a certain scale.

2) MSF Module: The Neck is an important component of the object detection framework. To achieve a better object detection result, researchers adopt the feature pyramid network FPN [35] modules to construct the Neck. The FPN which generally consists of three convolution blocks is a classic multiscale feature fusion module and has been widely used in many fields. In the FPN, the smaller scale feature maps will be upsampled to the same size as the larger scale feature maps, and both of them are fused together by pointwise addition to realize the cross-scale feature fusion. Based on the previous studies, we proposed an MSF module including two FPNs to realize multiscale feature map fusion step by step. The MSF adopts the multiscale feature maps f_1^{fpf}, f_2^{fpf}, and f_3^{fpf} as the input coming from FPF. The feature maps f_1^{fpf}, f_2^{fpf}, and f_3^{fpf} are first split at a certain proportion along the channel dimension to generate two sets of feature maps (f_{1-1}^{fpf}, f_{2-1}^{fpf}, f_{3-1}^{fpf}) and (f_{1-2}^{fpf}, f_{2-2}^{fpf}, f_{3-2}^{fpf}), with 16 channels and 112 channels, respectively. In the first step of feature fusion, the set (f_{1-1}^{fpf}, f_{2-1}^{fpf}, f_{3-1}^{fpf}) is processed by the first FPN to generate multiscale fusing results (f_{1-1}^{mfp}, f_{2-1}^{mfp}, f_{3-1}^{mfp}). The f_{1-1}^{mfp} which has the largest scale is used to predict the boxing mask by a convolution layer. Due to the supervision of boxing mask labels and multiscale feature fusion, the detective regions in the feature maps (f_{1-1}^{mfp}, f_{2-1}^{mfp}, f_{3-1}^{mfp}) can gain higher information intensity.

In the second step of feature fusion, the sets (f_{1-1}^{mfp}, f_{2-1}^{mfp}, f_{3-1}^{mfp}) and (f_{1-2}^{fpf}, f_{2-2}^{fpf}, f_{3-2}^{fpf}) are concatenated in the channel dimension according to different scales, respectively. The combined feature maps with channel number restored to 128 are handled by the second FPN to realize the full-data feature fusion, and the fused result is then sent to the Header for defect detection. With the guidance of the first fusion results (f_{1-1}^{mfp}, f_{2-1}^{mfp}, f_{3-1}^{mfp}) having higher information intensity in defect areas, the second FPN could pay more attention to the defects while processing the combined feature maps, and therefore generate a better multiscale feature fusion result. It should be noted that only a few feature maps are used to predict the boxing masks, which could therefore cause a bad prediction result due to insufficient defect information. However, the addition of the FPF between BoneNet and Neck can help the first FPN to predict a better boxing mask before the final defect detection.

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IV. Experiments

A. Evaluation Metrics

The mAP50, mAP75, mAP50:95, and mAR are most commonly used metrics to comprehensively evaluate the methods. Based on the confidence, category, and intersect over union (IOU) between the predicted bounding box and labeled bounding box, all the predicted results are divided into true positive (TP), false positive (FP), true negative (TN), and false negative (FN). The mean average precision (mAP) and mean average recall (mAR) can be calculated as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{3}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{4}$$

$$\mathrm{AP} = \int_{\mathrm{iou}_1}^{\mathrm{iou}_2} \int_0^1 P(\mathrm{th}_{\mathrm{iou}}, \mathrm{th}_{\mathrm{conf}}) \underset{R(\mathrm{th}_{\mathrm{iou}}, \mathrm{th}_{\mathrm{conf}})}{} dR \tag{5}$$

$$\mathrm{AR} = 2\int_{0.5}^{1} \mathrm{recall}(o)do \tag{6}$$

$$\mathrm{mAP} = \frac{1}{C}\sum_{i=1}^{C} \mathrm{AP}_i \tag{7}$$

$$\mathrm{mAR} = \frac{1}{C}\sum_{i=1}^{C} \mathrm{AR}_i \tag{8}$$

where iou_1 and iou_2 represent the lower and upper limits of IOU threshold, respectively, o represents the IOU overlap, and C represents the number of categories.

During the process of method evaluation, the mAP comprehensively considers the accuracy and recall of the method, reflecting the balance between the correctness and completeness of the method in determining positive samples. And the mAR mainly focuses on the recall ability of the method under different detection thresholds, reflecting the comprehensive ability of the model to recall positive samples.

When calculating the mAP, different values of iou_1 and iou_2 lead to different results. Therefore, we can obtain mAP50, mAP75, and mAP50:95 with their (iou_1, iou_2) being set to (50, 50), (75, 75), and (50, 95), respectively. Especially, the mAP50:95 is also called mAP. Then, the method's effectiveness under varying matching requirements is reflected by adopting distinct IOU thresholds.

Due to the particularity of industrial production, defect detection cares more about the false alarm and miss rate, which are, respectively, defined as

$$\mathrm{MR} = \frac{\mathrm{FN}}{\mathrm{FN}+\mathrm{TP}} \tag{9}$$

$$\mathrm{FA} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{10}$$

where MR is the miss rate, and FA is the false alarm. The same as the metrics precision and recall used in object detection, the miss rate and false alarm are also influenced by the threshold. For a comprehensive evaluation comparison on different algorithms, we adopt the false alarm and miss rate (F–M) curve with the area under the curve (AUC).

TABLE I

Performance Comparison of SYOLOv4 and FMN Trained on CD3 and CD3_En Datasets, Respectively

Model	Dataset	mAP50	mAP75	mAP	mAR
SYOLOv4	CD3	0.5064	0.2462	0.2636	0.4761
SYOLOv4	CD3_En	0.6484	0.3697	0.3754	0.5589
Improvement		0.1420↑	0.1235↑	0.1118↑	0.0828↑
FMN	CD3	0.5829	0.2689	0.3040	0.4677
FMN	CD3_En	0.7229	0.4129	0.4315	0.5886
Improvement		0.1400↑	0.1440↑	0.1275↑	0.1209↑

TABLE II

Performance Comparison of SYOLOv4 and FMN Trained on CD3 and CD3_En Datasets for Single-Class Defects, Respectively

Model	Dataset	mAP				mAR			
		RC	crush	grain	scratch	RC	crush	grain	scratch
SYOLOv4	CD3	0.5625	0.2429	0.1015	0.1473	0.7042	0.5027	0.2659	0.4316
SYOLOv4	CD3_En	0.7589	0.3922	0.1370	0.2136	0.8375	0.5526	0.3462	0.4991
Improvement		0.1964↑	0.1493↑	0.0355↑	0.0663↑	0.1333↑	0.0499↑	0.0803↑	0.0675↑
FMN	CD3	0.6267	0.2838	0.1055	0.2001	0.7000	0.4500	0.2647	0.4560
FMN	CD3_En	0.8291	0.4481	0.1594	0.2895	0.8625	0.5711	0.3626	0.5581
Improvement		0.2024↑	0.1643↑	0.0539↑	0.0894↑	0.1625↑	0.1211↑	0.0979↑	0.1021↑

TABLE III

Ablation Studies of the Proposed Modules. The Best Scores Are Marked in Bold

No.	FPF	MSF	mask	mAP50	mAP75	mAP	mAR
M1				0.6484	0.3697	0.3754	0.5589
M2	✓			0.6730	0.4036	0.4102	0.5793
M3		✓		0.6608	0.3795	0.3914	0.5586
M4	✓	✓		0.7029	0.3952	0.4179	0.5739
M5		✓	✓	0.6730	0.4112	0.4065	0.5867
M6	✓	✓	✓	**0.7229**	**0.4129**	**0.4315**	**0.5886**

B. Implementation Details

When we construct datasets CD3 and CD3_En, a sliding window with size of 150 × 150 pixels is used to generate subimages, and the sliding step is set to half of the window's size. The input image is resized to 352 × 352 for training and predicting. The maximum learning rate is set to 0.001 and the cosine strategy is used to adjust the learning rate during the training process. The Adam optimizer with L2 regularization is selected to optimize the parameters. The batch size and the maximum epoch are set to 20 and 30, respectively. We use PyTorch 1.10 to implement the FMN with two RTX2080Ti GPUs to accelerate model optimization.

C. Comparison of Dataset Enhancement

Compared with the CD3 dataset whose training set has 7475 samples, the CD3_En dataset enhanced by the defect copy method and background reuse method in sequence has 37 707 samples in the training set. Sufficient data in CD3_En can strongly support model optimization.

The different probabilities of four defects in camera decorations lead to severe category imbalance problem. The training set in the CD3 contains 68 RC defects, 191 crush defects, 523 scratch defects, and 11 954 grain defects; the training set in CD3_En contains 5525 RC defects, 5811 crush defects, 9625 scratch defects, and 41 924 grain defects. The ratio between the most numerous defects (grain) and the fewest numerous defects (RC) decreases from 175.8 for CD3 to 7.6 for CD3_En, which provides a convincingly support that the category imbalance issues have been alleviated.

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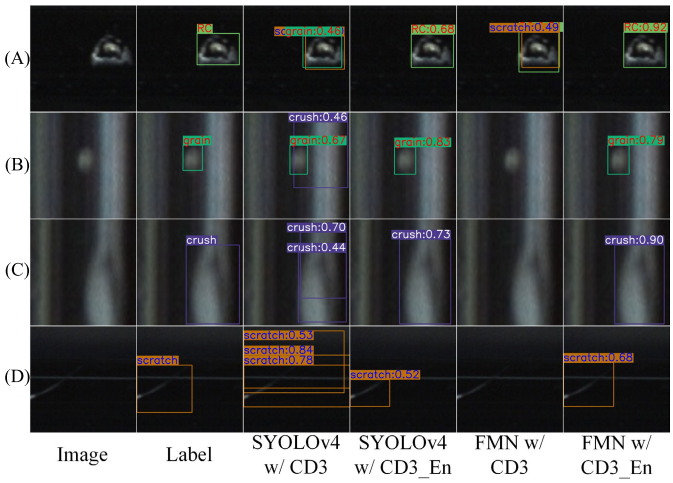


Fig. 7. Visual comparison of defect detection results of SYOLOv4 and FMN trained on CD3 and CD3_En, respectively. Samples (A)–(D) are RC, grain, crush, and scratch defects, respectively.

To verify the superiority of the enhanced dataset CD3_En, we perform the training and testing processes on CD3 and CD3_En, respectively. The persuasiveness of the experiment is worth ensuring as the test sets in these two datasets are the same. We use two algorithms, SYOLOv4 [31] and the proposed FMN method, to prove the advantage of the enhanced dataset CD3_En, and the results are shown in Table I. The SYOLOv4 trained on CD3_En performs better than trained on CD3 as the mAP50, mAP75, mAP, and mAR improve by 14.2%, 12.35%, 11.18%, and 8.28%, respectively, and for FMN, these metrics improve by 14.00%, 14.40%, 12.75%, and 12.09%, respectively.

We also calculate the mAP and mAR metrics for single class to further validate the superiority of the dataset enhancing framework, as shown in Table II, from which we can find that compared with the testing results generated with CD3-trained models, the models trained on CD3_En demonstrate better performance on all eight metrics.

We selected four typical samples for their visual comparison on detection results, as shown in Fig. 7. For sample (A) containing an RC defect, the SYOLOv4 and FMN methods trained on CD3 false detect target as scratch defect, while those trained on CD3_En provide a correct detection result. The FFM trained on CD3 encounters missing defects on samples (B)–(D), while that trained on CD3_En presents satisfactory detection results. The SYOLOv4 trained on CD3 has false detects on samples (B)–(D), while that trained on CD3_En has correct results. These visualization results demonstrate the superiority of the defect copy method and background reuse method on dataset enhancement for camera decoration surface defect detection.

D. Ablation Studies of the Proposed Modules

In this section, we conduct a series of controlled experiments based on the SYOLOv4 which is regarded as the baseline to investigate the effectiveness of the devised modules. All the experiments are conducted on the enhanced dataset CD3_En and use mAP50, mAP75, mAP, and mAR to measure their performances.

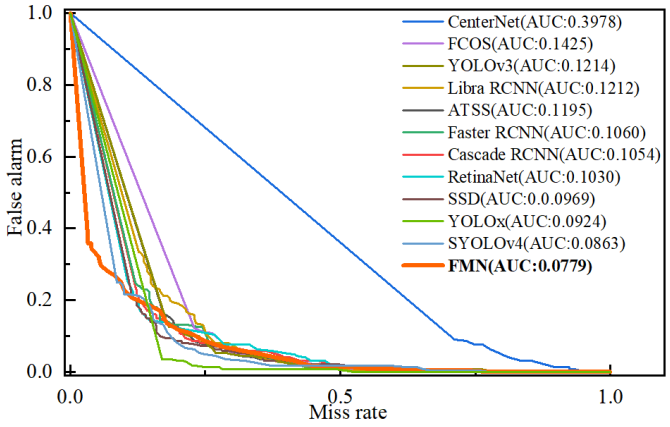


Fig. 8. Comparisons of F–M curves and AUCs on FMN and the other 11 classic methods. The smaller the AUC, the better the algorithm.

TABLE IV
PERFORMANCE COMPARISONS OF FMN AND 11 CLASSIC METHODS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	BoneNet	mAP50	mAP75	mAP	mAR
ATSS	ResNet50	0.684	0.399	0.402	0.528
Cascade RCNN	ResNet50	0.677	**0.419**	0.406	0.49
CenterNet	ResNet50-DCN	0.676	0.414	0.395	0.489
Faster RCNN	ResNet50	0.691	0.394	0.402	0.495
FCOS	ResNet50	0.676	0.388	0.393	0.519
Libra RCNN	ResNet50	0.658	0.375	0.387	0.498
RetinaNet	ResNet50	0.676	0.352	0.391	0.508
SSD	VGG16	0.658	0.353	0.381	0.48
YOLOv3	DarkNet53	0.673	0.406	0.389	0.489
YOLOx	DarkNet53-CSP	0.668	0.414	0.415	0.525
SYOLOv4	DarkNet53-CSP	0.6484	0.3697	0.3754	0.5589
FMN(Ours)	DarkNet53-CSP	**0.7229**	0.4129	**0.4315**	**0.5886**

Table III summarizes the experimental results used to assess the effectiveness of the designed modules, including FPF, MSF, and mask used in MSF. Compared with the baseline method M1, both M2 adopting FPF and M3 adopting MSF promote the four metrics obviously. The performances of M2 and M3 demonstrate the superiority of FPF and MSF which provide information penetration and two-step feature fusion, respectively. Meanwhile, M4 where FPF and MSF are added improves the metrics mAP50, mAP75, mAP, and mAR by 5.45%, 2.55%, 4.25%, and 1.5%, respectively. The improvement of four metrics in M4 further verifies that FPF and MSF are complementary.

Different from the FPF and MSF which could be used independently, the mask used as the guidance must rely on the MSF module. Therefore, we supply the experiments of M5 and M6 which are separately based on M3 and M4 to verify the effectiveness of mask guidance. Compared with M3, the performance of M5 using mask guidance in MSF improves by 1.22%, 3.17%, 1.51%, and 2.81% on mAP50, mAP75, mAP, and mAR, respectively. Moreover, the proposed FMN where mask guidance is added into M4 presents the best performance, as shown by M6 in Table III. The above comparisons demonstrate the effectiveness of mask guidance.

E. Comparison With Classic Methods

1) Quantitative Comparison: We train and test the proposed FMN and other 11 classic object detection methods on CD3_En including ATSS [36], Cascade RCNN [37], CenterNet [38], Faster RCNN [28], FCOS [39], Libra RCNN

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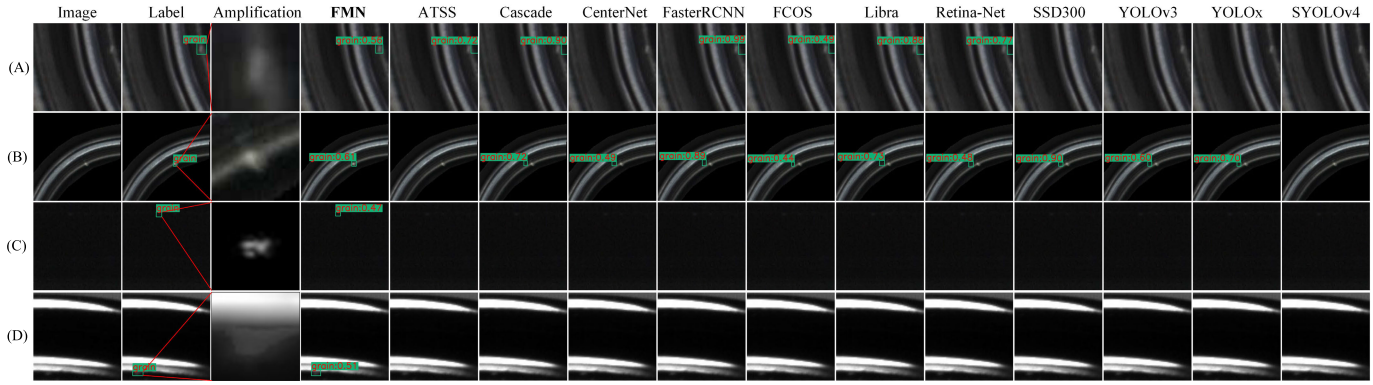


Fig. 9. Visual comparison of defect detection results among the FMN and the other 11 classic methods. Row (A) is a sample containing an inconspicuous defect similar to the background; row (B) is tiny defect; rows (C) and (D) are tiny and inconspicuous defects. For a clearer observation, the amplification in rows (C) and (D) has enhanced contrast.

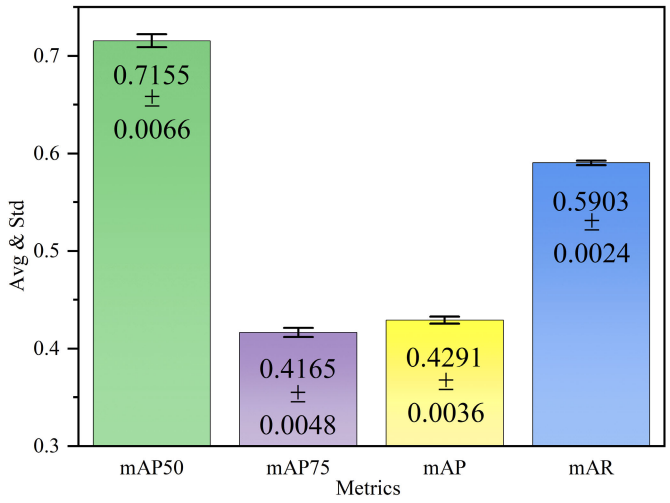


Fig. 10. Mean and standard deviation of evaluation metrics for three trained FMN models.

[40], RetinaNet [41], SSD [26], YOLOv3 [42], YOLOx [32], and SYOLOv4 [31]. For a fair comparison, all these methods are trained with the strategies provided by the authors and we use the same metrics to measure the performances of different methods, as shown in Table IV. Compared with the best performance of 11 classic methods, the proposed FMN improves by 3.19%, 1.65%, and 2.97% on mAP50, mAP, and mAR, respectively, which proves its superior in camera decoration defect detection. However, it is worth noting that the Cascade RCNN provides the best performance in mAP75. We attribute it to a cascade head structure which provides better performance at higher threshold by gradually increasing the IOU threshold. Unfortunately, this strategy makes the method perform worse in mAP50 since too less attention is paid to head with lower threshold. We also provide the F–M curves with the AUCs for each method, as shown in Fig. 8. The FMN achieves the smallest AUC among these trained methods, which further demonstrates the superiority of the proposed method on camera decoration defect detection.

2) Qualitative Comparison: To qualitatively compare and evaluate the performance of the proposed FMN and the other 11 classic methods, we visualize some defect detection results of challenging scenes, as displayed in Fig. 9. For the defects that are inconspicuous or tiny as in rows (A) and (B), the 11 classic methods generate the results with location deviation which can be attributed to the ability deficiency of dealing tiny object as the contribution of Euclidean location deviation loss to training is minimal. And this issue is addressed in our method by the introducing of boxing mask prediction which facilitates method to be more sensitive to location. While for the defects that are inconspicuous and tiny as in rows (C) and (D), the 11 classic methods present bad detection results with low confidence which are filtered out by threshold. It is obvious that our proposed method performs better in various difficult scenes, which is mainly because the developed FPF module can provide information permeation for context enhancement among multiscale feature maps and the MSF with mask guidance can generate high-quality feature maps in which the defect areas have higher information intensity.

F. Stability Assessment

To further verify the stability of the proposed framework FMN, two other FMN models are trained and tested. We calculate the mean and standard deviation of evaluation metrics for three trained FMN models, as illustrated in Fig. 10. The standard deviation of mAP50, mAP75, mAP, and mAR is 0.0066, 0.0048, 0.0036, and 0.0024, respectively. It is obviously that the proposed framework FMN has tiny fluctuation on four metrics, which provides sufficient evidence that our framework has excellent stability.

G. Failure Case Analysis

Although the proposed FMN is good at dealing challenging scenes, it may still fail in some cases, as illustrated in Fig. 11. In Fig. 11(A), the presented 12 results omit the real defect due to the extremely week contrast between the defect and the background. Moreover, the proposed FMN and the Libra misjudges a reflection as a grain defect. Therefore, the ability of identifying extremely inconspicuous defect and resisting the interference of reflection should be enhanced. In Fig. 11(B) and (C), the defects are tenuous and the middle part of defects is inconspicuous. Thus, all 12 methods fail to identify the intact defects. Some methods regard the partial areas with distinct feature as the grain defects. Other methods perform worse as they omit the defects. It is worth noting that the Libra generates a bounding box containing the whole defect, but expands the area of defect. To address this issue, establishing strong correlations within a single defect and

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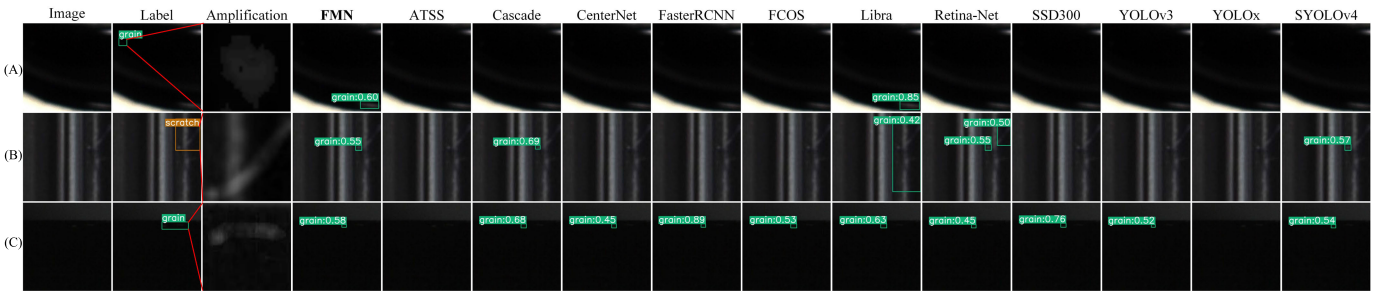


Fig. 11. Visual comparison of failure cases among the FMN and the other 11 classic methods, where rows (A)–(C) display different cases. For a clearer observation, amplification in rows (A) and (C) has enhanced contrast.

enhancing the integrity of detection results need to be further explored.

V. Conclusion

In this article, we construct a camera decoration defect dataset named CD3 and propose an efficient dataset enhancing framework including the defect copy method and background reuse method to solve the problems of insufficient samples and imbalanced categories. We also provide an enhanced camera decoration defect dataset named CD3_En containing 37 707 image samples in the training set and 1942 ones in the test set. Moreover, an effective framework FMN mainly consisting of FPF, MSF, and mask guidance is present to detect camera decoration surface defect. In the experimental studies, we provide four comparison models to prove the efficiency of the proposed dataset enhancing framework and train six comparison models to demonstrate the effectiveness of the proposed modules. The brilliant performances in quantitative and qualitative comparisons prove the robustness and effectiveness of the proposed FMN which has the potential to be used for defect detection of various challenging scenarios.

Acknowledgment

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Hui Wang received B.S. degree from the School of Automation, Central South University, Changsha, China, in 2021, where he is currently pursuing the Ph.D. degree.

His research interests include salient object detection, object detection, computer vision, and deep learning.



Yuqian Zhao received the M.Eng. degree in automatic control engineering and the Ph.D. degree in computer science and technology from Central South University, Changsha, China, in 2002 and 2006, respectively.

He was engaged in post-doctoral research at New Jersey Institute of Technology, Newark, NJ, USA, from 2009 to 2010. He is currently a Full Professor with Central South University. His research interests include image processing, pattern recognition, computer vision, video analysis, and intelligent sensing.



Fan Zhang received the Ph.D. degree from Central South University, Changsha, China, in 2019.

He is currently an Associate Professor with Central South University. His research interests include image processing, computer vision, and pattern recognition.



Gui Gui (Member, IEEE) received the M.Eng. degree in computer science from Central South University, Changsha, China, in 2003, and the Ph.D. degree in computer science from University of Essex, Colchester, U.K., in 2007.

She is currently a Full Professor with the School of Automation, Central South University. Her current research interests include artificial intelligence, data modeling, and big data system.



Qiwu Luo (Senior Member, IEEE) received the B.S. degree in communication engineering from the National University of Defense Technology, Changsha, China, in 2008, and the M.Sc. degree in electronic science and technology and the Ph.D. degree in electrical engineering from Hunan University, Changsha, in 2011 and 2016, respectively.

He served as a Senior Engineer of instrumentation with WASION Group Company Ltd., Changsha, and the Deputy Technical Director with Hunan RAMON Technology Company Ltd., Changsha. In 2016, he worked at the School of Electrical Engineering and Automation, Hefei University of Technology, Hefei, China, where he also completed his post-doctoral research on automatic optic inspection (AOI). He is currently a Full Professor with the School of Automation, Central South University, Changsha, where he conducts research in computer vision, industrial AOI, optical instrumental measurement, parallel hardware architecture design, and reconfigurable computing.



Chunhua Yang (Fellow, IEEE) received the M.S. degree in automatic control engineering and the Ph.D. degree in control science and engineering from Central South University, Changsha, China, in 1988 and 2002, respectively.

From 1999 to 2001, she was a Visiting Professor with the University of Leuven, Leuven, Belgium. Since 1999, she has been a Full Professor with the School of Information Science and Engineering, Central South University, Changsha. From 2009 to 2010, she was a Senior Visiting Scholar with the University of Western Ontario, London, ON, Canada. She is currently the HoD of the School of Automation, Central South University. Her current research interests include modeling and optimal control of complex industrial processes, and intelligent control systems.



Weihua Gui (Member, IEEE) received the B.Eng. degree in electrical engineering from Central South University, Changsha, China, in 1976.

From 1986 to 1988, he was a Visiting Scholar with the University of Duisburg-Essen, Duisburg, Germany. Since 1991, he has been a Professor with the School of Automation, Central South University. Since 2013, he has been an Academician with the Chinese Academy of Engineering, Beijing, China. His research interests include modeling and optimal control of complex industrial processes, distributed robust control, and fault diagnosis.

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